Analysis Pilot 2

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Load Packages

```
library(tidyverse)
```

```
## Warning: Paket 'ggplot2' wurde unter R Version 4.4.3 erstellt
```

```
## Warning: Paket 'stringr' wurde unter R Version 4.4.3 erstellt
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.6.0
## v ggplot2   4.0.0     v tibble    3.2.1
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.0.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(magrittr)
```

```
##
## Attache Paket: 'magrittr'
##
## Das folgende Objekt ist maskiert 'package:purrr':
##
##     set_names
##
## Das folgende Objekt ist maskiert 'package:tidyr':
##
##     extract
```

```
library(lme4)
```

```
## Lade nötiges Paket: Matrix
##
## Attache Paket: 'Matrix'
##
## Die folgenden Objekte sind maskiert von 'package:tidyr':
##
##     expand, pack, unpack
```

```r
library(lmerTest)
```

```
##
## Attache Paket: 'lmerTest'
##
## Das folgende Objekt ist maskiert 'package:lme4':
##
##     lmer
##
## Das folgende Objekt ist maskiert 'package:stats':
##
##     step
```

```r
library(here)
```

```
## here() starts at C:/Users/mas48yu/Desktop/kaffklatsch
```

```r
library(performance)
```

Import and Rename Data, Initial Exclusions

Why name this differently?

```r
# load final data file!
raw_data <- read_csv(here("pilot2/data_pilot2_all.csv"))
```

```
## Rows: 448 Columns: 516
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr  (228): QUESTNNR, MODE, DE05_03, DE09_02, DE10_01, IV01_01, IV01_02, IV0...
## dbl  (283): CASE, AC01, AC02, AC03, AC04, AC05, AC06, DE02_01, DE03, DE04, D...
## lgl    (3): SERIAL, REF, MAILSENT
## dttm   (2): STARTED, LASTDATA
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

Change values -9 to NA

```r
raw_data[raw_data == -9] <- NA
```

Rename variables

```r
raw_data %<>%
  rename(age = DE02_01,
         education = DE03,
         field.of.study = DE04,
         gender = DE05,
         exclusion = DE09,
```

```
        exclusion.reasons = DE09_02,
        comments = DE10_01,
        data.usage = DE11,
        consent = IC02,
        device = PI01_FmF,
        screen.width= PI01_ScW,
        screen.height = PI01_ScH,
        prolificID = PI01_RV1,
        sessionID = PI01_RV2,
        studyID = PI01_RV3,
        quality.indicators.new1 = QI02x01,
        quality.indicators.new2 = QI02x02,
        quality.indicators.new3 = QI02x03,
        quality.indicators.new4 = QI02x04) %>%

rename_with(.cols = starts_with("AC"),
            .fn = ~paste0("attention.check_", 1:6)) %>%

rename_with(.cols = starts_with("IV01"),
            .fn = ~paste0("vignette_", 1:24)) %>%

rename_with(.cols = starts_with("IV02"),
            .fn = ~paste0("title_", 1:24)) %>%

rename_with(.cols = starts_with("IV03"),
            .fn = ~paste0("name_", 1:24)) %>%

rename_with(.cols = starts_with("IV04"),
            .fn = ~paste0("institution_", 1:24)) %>%

rename_with(.cols = starts_with("IV05"),
            .fn = ~paste0("confirmation_", 1:24)) %>%

rename_with(.cols = starts_with("IV06"),
            .fn = ~paste0("confirmation.text_", 1:24)) %>%

rename_with(.cols = starts_with("IV07"),
            .fn = ~paste0("quality_", 1:24)) %>%

rename_with(.cols = starts_with("IV08"),
            .fn = ~paste0("quality.text_", 1:24)) %>%

rename_with(.cols = starts_with("IV09"),
            .fn = ~paste0("prestige_", 1:24)) %>%

rename_with(.cols = starts_with("IV10"),
            .fn = ~paste0("prestige.text_", 1:24)) %>%

rename_with(.cols = starts_with("IV11"),
            .fn = ~paste0("opendata_", 1:24)) %>%

rename_with(.cols = starts_with("IV12"),
            .fn = ~paste0("opendata.text_", 1:24)) %>%
```

```
  rename_with(.cols = starts_with("IV13"),
              .fn = ~paste0("citation_", 1:24)) %>%

  rename_with(.cols = starts_with("IV14"),
              .fn = ~paste0("citation.text_", 1:24)) %>%

  rename_with(.cols = starts_with("QI01"),
              .fn = ~paste0("quality.indicator_", 1:6)) %>%

  rename_with(.cols = starts_with("QR"),
              .fn = ~paste0("quality.rating_", 1:24)) %>%

  rename_with(.cols = starts_with("SO"),
              .fn = ~paste0("category_", 1:24))
```

Data exclusions

Mark participants which may have participated several times (except if first time was completed)

```
exclusions <- raw_data %>%
  #participants are allowed to read the instructions twice but not pages with manipulation
  #thus, only inspect trials where participants got to page four or above
  filter(MAXPAGE > 3) %>%

  # filter for duplicated prolific ids
  group_by(prolificID) %>%
  filter(n() > 1) %>%

  # get participants who completed on the second or later try
  # incomplete tries are excluded anyway
  arrange(STARTED) %>%
  filter(STARTED != min(STARTED) & MAXPAGE >= 11) %>%
  ungroup() %>%

  # get reason for exclusion and save IDs for exclusion
  mutate(reason = "not completed on first try") %>%
  select(CASE, reason)
```

Check if there are participants that retracted consent

```
exclusions %<>%
  bind_rows(raw_data %>%
              filter(data.usage == 2) %>%
              mutate(reason = "retracted consent") %>%
              select(CASE, reason))
```

Exclude 3 participants: CASE 142 (technical problems) and CASES 485 and 510 (could not focus)

```
exclusions %<>%
  bind_rows(raw_data %>%
              filter(CASE %in% c(142, 485, 510)) %>%
```

```
              mutate(reason = "technical issues") %>%
              select(CASE, reason))
```

Compute window change indicator TODO: Do we plan exclusions based on this?

```
raw_data <- raw_data %>%
  rowwise() %>%
  mutate(window.changes = sum(c_across(WC01_01 : WC03_24))) %>%
  ungroup()
```

remove participants who meet exclusion criteria

```
raw_data <- filter(raw_data, !CASE %in% exclusions$CASE)
```

Filter for participants to have reached last page and answered last question (final consent): complete cases

```
raw_data <- filter(raw_data, MAXPAGE >= 11, !is.na(data.usage))
```

Remove prolific IDs

```
raw_data$prolificID <- NA
```

Data Wrangling

Select and extract relevant variables, convert to long format

```
data_long <- raw_data %>%
  select(CASE, starts_with("attention.check"), vignette_1:citation.text_24, starts_with("quality.rating
  pivot_longer(cols = attention.check_1:quality.rating_24,
               names_to = c(".value", "item"),
               names_sep = "_")
```

Evaluate attention checks - exclude if less than 4 passed

```
exclusions %<>%
  bind_rows(
    data_long %>%
      filter(item %in% c(1:6)) %>%
      # recode attention checks
      mutate(attention.check = ifelse(item %in% c(2,3,5,6) & attention.check == 2, 0, attention.check),
             ac.passed = case_when(item %in% c(1,4) ~ attention.check == 1,
                                   item %in% c(2,5) ~ attention.check == confirmation,
                                   item %in% c(3,6) ~ attention.check == quality)) %>%
      group_by(CASE) %>%
      summarise(ac.passed = sum(ac.passed)) %>%
      filter(ac.passed < 4) %>%
      mutate(reason = "attention check failed") %>%
      select(CASE, reason))
```

Exclude if attention check failed

```r
raw_data <- filter(raw_data, !CASE %in% exclusions$CASE)
data_long <- filter(data_long, !CASE %in% exclusions$CASE)
```

Total number of excluded participants

```r
length(exclusions$CASE)
```

```
## [1] 89
```

Number of remaining participants

```r
length(unique(raw_data$CASE))
```

```
## [1] 210
```

Compute IV: Rename and code quality indicator variables

```r
data_long %<>%
  mutate(title_r = ifelse(title == "Prof.", 1, 0),
         institution_r = ifelse(institution %in% c("Harvard University",
                                                   "Stanford University",
                                                   "Cornell University",
                                                   "University of California, Berkeley",
                                                   "University of Michigan",
                                                   "University of Washington",
                                                   "Columbia University",
                                                   "University of Pennsylvania",
                                                   "Yale University",
                                                   "Princeton University"),
                                1, 0),
         vignette_r = ifelse(vignette %in% c("VI01", "VI02"), 1, 0)) #1 = long vignette, 0 = short vign
```

Compute DV: Filter out missing quality ratings and recode

```r
data_long <- data_long %>%
  filter(!is.na(quality.rating)) %>%
  mutate(quality.rating = (quality.rating-1)/100)
```

Demographics

Age

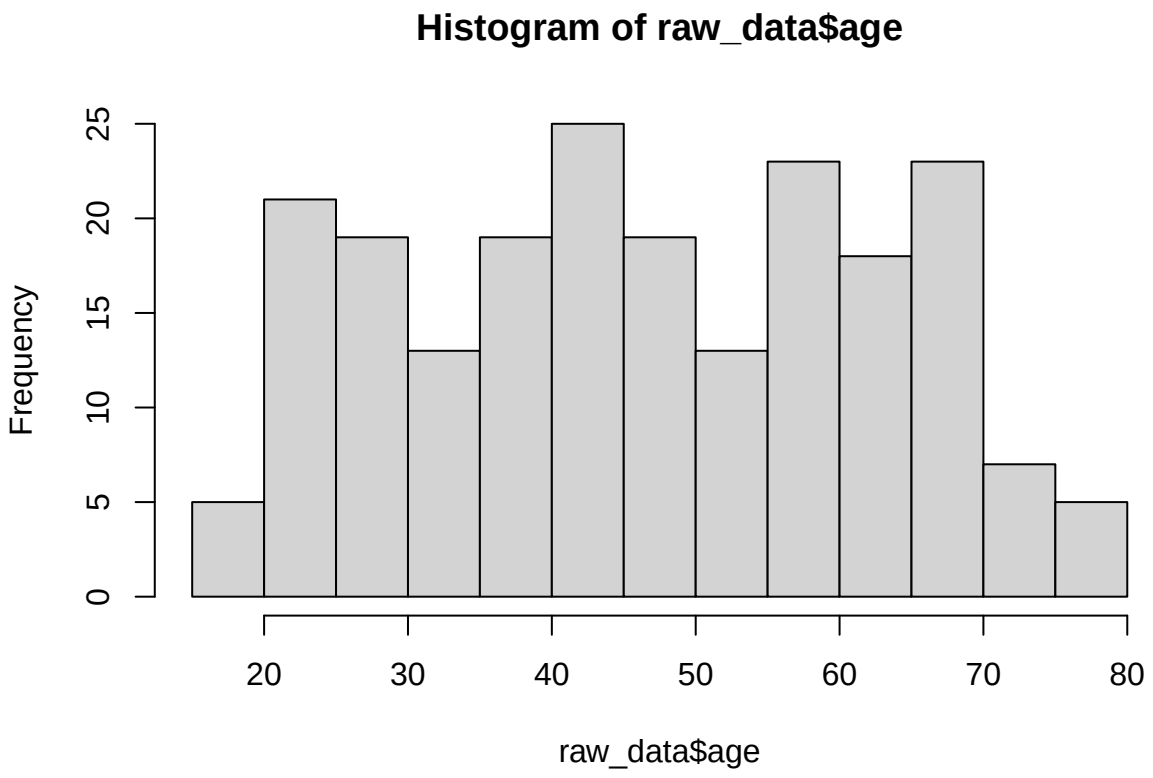
```r
mean(raw_data$age)
```

```
## [1] 46.90476
```

```r
sd(raw_data$age)
```

```
## [1] 16.11444
```

```r
hist(raw_data$age)
```



Gender

```r
table(raw_data$gender)
```

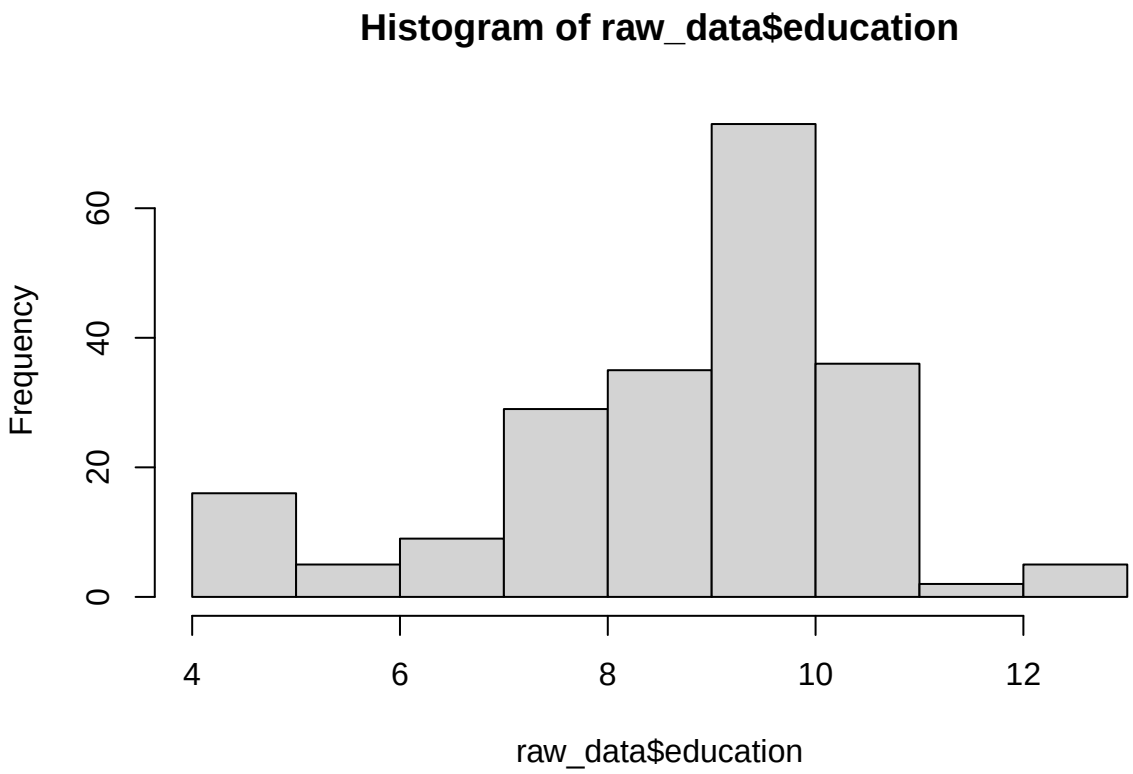
```
##
##   1   2   3
## 101 107   2
```

Education

```r
table(raw_data$education)
```

```
##
##  4  5  6  7  8  9 10 11 12 13
##  2 14  5  9 29 35 73 36  2  5
```

```r
hist(raw_data$education)
```



Field of study

```
table(raw_data$field.of.study)
```

```
##
##  1  2  3  4  5  6
##  9 57 19  2 48 40
```

Analysis

Descriptives

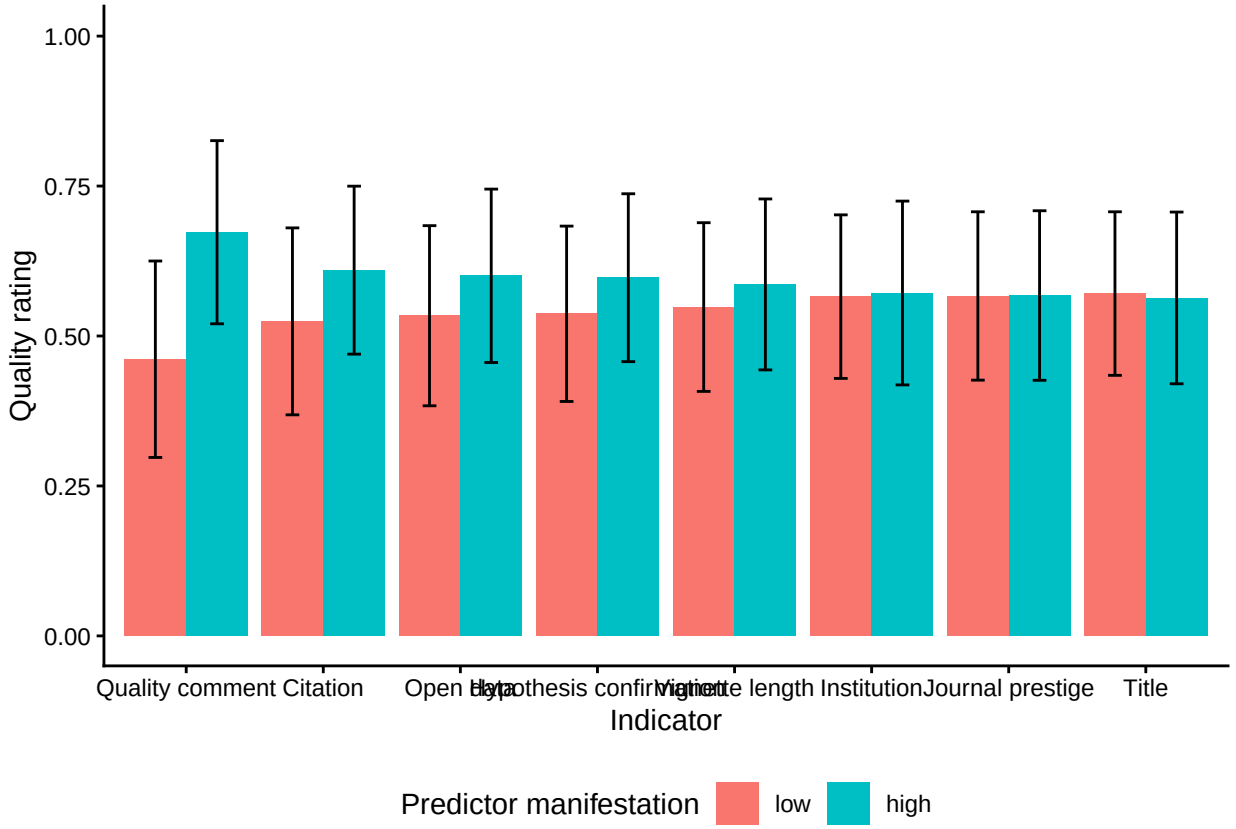
```
desc <- data_long %>%
  select(CASE, vignette, quality.rating, title_r, institution_r, confirmation, quality, prestige, openda
  pivot_longer(cols = title_r:vignette_r, names_to = "indicator", values_to = "indicator_value")

desc_avg <- desc %>%
  group_by(CASE, indicator, indicator_value) %>%
  summarise(qr_ind = mean(quality.rating)) %>%
  group_by(indicator, indicator_value) %>%
  summarise(qr = mean(qr_ind), se = sd(qr_ind)) %>%
  group_by(indicator) %>%
  mutate(diff = qr - lag(qr)) %>%
  mutate(diff = if_else(is.na(diff), lead(diff), diff)) %>%
  ungroup() %>%
  arrange(desc(diff)) %>%
  mutate(indicator_value = if_else(indicator_value == 0, "low", "high")) %>%
```

```
  mutate(indicator_value = factor(indicator_value, levels = c("low", "high")),
         indicator = factor(indicator,
                            levels = unique(indicator),
                            labels = c("Quality comment", "Citation", "Open data", "Hypothesis confirma
```

```
## `summarise()` has grouped output by 'CASE', 'indicator'. You can override using
## the `.groups` argument.
## `summarise()` has grouped output by 'indicator'. You can override using the
## `.groups` argument.
```

```
p_qr <- ggplot(data = desc_avg, mapping = aes(x = indicator, y = qr, fill = indicator_value, group = in
  geom_col(position = position_dodge(.9)) +
  geom_errorbar(mapping = aes(ymin = qr-se, ymax = qr+se), width = .2, position = position_dodge(.9)) +
  theme_classic() +
  ylim(c(0,1)) +
  labs(y = "Quality rating",
       x = "Indicator",
       fill = "Predictor manifestation") +
  theme(legend.position = "bottom")
p_qr
```



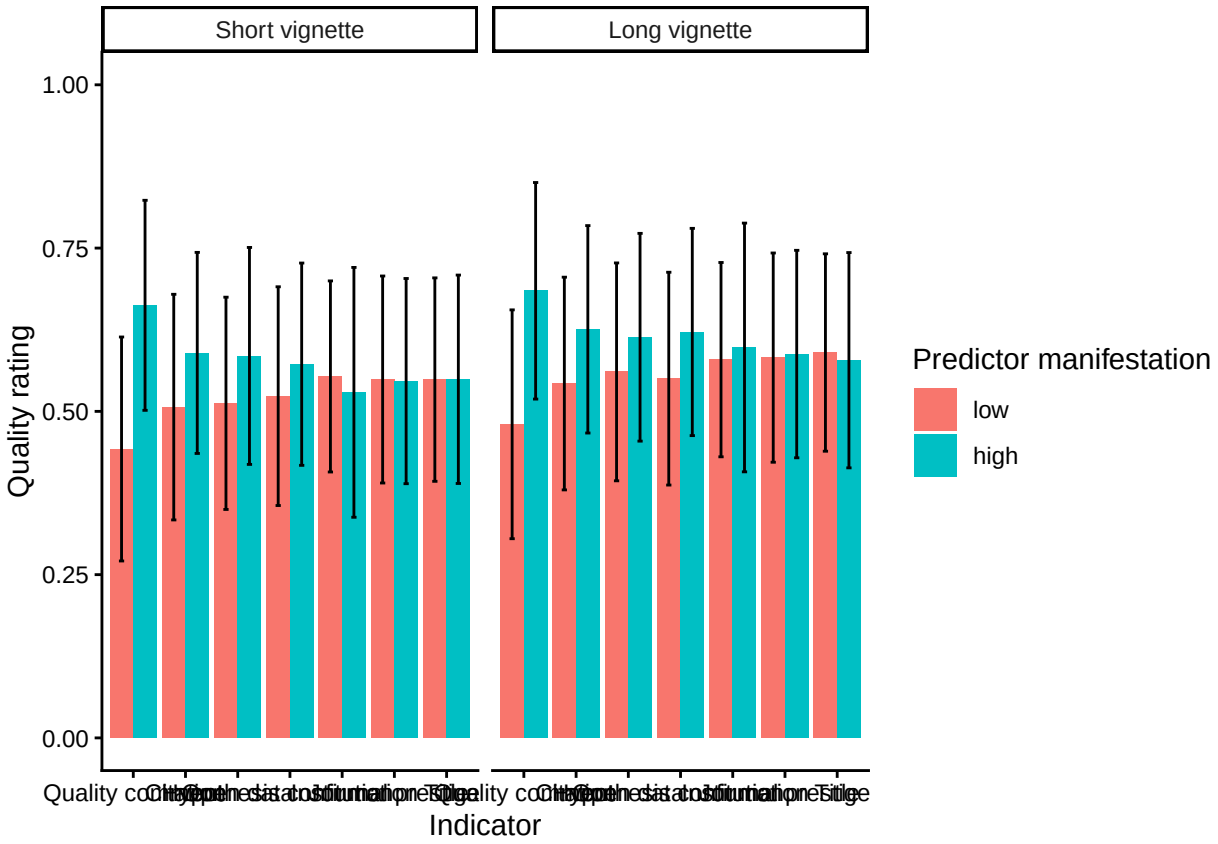

```
# ggsave(p_qr, file = here("pilot2/plot_qr.svg"), height = 10, width = 15, limitsize = FALSE)
```

```r
desc2 <- data_long %>%
  select(CASE, vignette, quality.rating, title_r, institution_r, confirmation, quality, prestige, opend
  pivot_longer(cols = title_r:citation, names_to = "indicator", values_to = "indicator_value")

desc2_avg <- desc2 %>%
  group_by(CASE, indicator, indicator_value, vignette_r) %>%
  summarise(qr_ind = mean(quality.rating)) %>%
  group_by(indicator, indicator_value, vignette_r) %>%
  summarise(qr = mean(qr_ind), se = sd(qr_ind)) %>%
  mutate(indicator_value = if_else(indicator_value == 0, "low", "high")) %>%
  mutate(indicator_value = factor(indicator_value, levels = c("low", "high")),
         indicator = factor(indicator,
                            levels = c("quality", "citation", "opendata", "confirmation", "institution_
                            labels = c("Quality comment", "Citation", "Open data", "Hypothesis confirma
        vignette_r = factor(vignette_r, labels = c("Short vignette", "Long vignette")))
```

```
## `summarise()` has grouped output by 'CASE', 'indicator', 'indicator_value'. You
## can override using the `.groups` argument.
## `summarise()` has grouped output by 'indicator', 'indicator_value'. You can
## override using the `.groups` argument.
```

```r
ggplot(data = desc2_avg, mapping = aes(x = indicator, y = qr, fill = indicator_value, group = indicator
  geom_col(position = position_dodge(.9)) +
  geom_errorbar(mapping = aes(ymin = qr-se, ymax = qr+se), width = .2, position = position_dodge(.9)) +
  theme_classic() +
  facet_wrap(~vignette_r) +
  ylim(c(0,1)) +
  labs(y = "Quality rating",
       x = "Indicator",
       fill = "Predictor manifestation")
```




```
qr_avg <- data_long %>%
  group_by(title_r, institution_r, confirmation, quality, prestige, opendata, citation) %>%
  summarise(qr = mean(quality.rating))
```

```
## `summarise()` has grouped output by 'title_r', 'institution_r', 'confirmation',
## 'quality', 'prestige', 'opendata'. You can override using the `.groups`
## argument.
```

Mixed linear model analysis

Effect coding

```
data_long$title_r <- data_long$title_r - 0.5
data_long$institution_r <- data_long$institution_r - 0.5
data_long$confirmation <- data_long$confirmation - 0.5
data_long$quality <- data_long$quality - 0.5
data_long$prestige <- data_long$prestige - 0.5
data_long$opendata <- data_long$opendata - 0.5
data_long$citation <- data_long$citation - 0.5
data_long$vignette_r <- data_long$vignette_r - 0.5
```

Fit model

```r
mod_inter <- lmer(quality.rating ~ title_r + institution_r + confirmation + quality + prestige + opendat
summary(mod_inter)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: quality.rating ~ title_r + institution_r + confirmation + quality +
##     prestige + opendata + citation + vignette_r + title_r:vignette_r +
##     institution_r:vignette_r + confirmation:vignette_r + quality:vignette_r +
##     prestige:vignette_r + opendata:vignette_r + citation:vignette_r +
##     (1 | CASE) + (1 | item)
##    Data: data_long
##
## REML criterion at convergence: -1342.6
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -3.3919 -0.6148  0.0472  0.6763  3.7541
##
## Random effects:
##  Groups   Name        Variance  Std.Dev.
##  CASE     (Intercept) 0.0158227 0.12579
##  item     (Intercept) 0.0003451 0.01858
##  Residual             0.0395849 0.19896
## Number of obs: 5035, groups:  CASE, 210; item, 24
##
## Fixed effects:
##                            Estimate Std. Error         df t value Pr(>|t|)
## (Intercept)               5.687e-01  1.001e-02  1.838e+02  56.795  < 2e-16 ***
## title_r                  -2.902e-03  5.622e-03  4.796e+03  -0.516   0.6058
## institution_r             5.562e-03  6.499e-03  4.800e+03   0.856   0.3921
## confirmation              5.702e-02  5.631e-03  4.806e+03  10.127  < 2e-16 ***
## quality                   2.104e-01  5.622e-03  4.801e+03  37.428  < 2e-16 ***
## prestige                  6.250e-03  5.629e-03  4.801e+03   1.110   0.2669
## opendata                  6.548e-02  5.637e-03  4.804e+03  11.617  < 2e-16 ***
## citation                  8.310e-02  5.628e-03  4.803e+03  14.764  < 2e-16 ***
## vignette_r                3.895e-02  6.526e-03  4.804e+03   5.968 2.57e-09 ***
## title_r:vignette_r       -4.363e-03  1.151e-02  4.850e+03  -0.379   0.7046
## institution_r:vignette_r  3.076e-02  1.325e-02  4.836e+03   2.321   0.0203 *
## confirmation:vignette_r   2.243e-02  1.147e-02  4.841e+03   1.956   0.0505 .
## quality:vignette_r       -1.943e-02  1.149e-02  4.844e+03  -1.690   0.0910 .
## prestige:vignette_r      -3.977e-03  1.147e-02  4.838e+03  -0.347   0.7288
## opendata:vignette_r      -1.778e-02  1.149e-02  4.837e+03  -1.548   0.1217
## citation:vignette_r      -2.415e-03  1.148e-02  4.838e+03  -0.210   0.8335
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

##
## Correlation matrix not shown by default, as p = 16 > 12.
## Use print(x, correlation=TRUE)  or
##     vcov(x)        if you need it
```

Waldian confidence intervals for fixed effects

```
CIWald <- confint(mod_inter, method = "Wald", oldNames = FALSE)
CIWald
```

```
##                                  2.5 %      97.5 %
## sd_(Intercept)|CASE                 NA          NA
## sd_(Intercept)|item                 NA          NA
## sigma                               NA          NA
## (Intercept)               5.490654e-01 0.588315853
## title_r                  -1.392053e-02 0.008116930
## institution_r            -7.176295e-03 0.018300562
## confirmation              4.598274e-02 0.068054348
## quality                   1.994163e-01 0.221455799
## prestige                 -4.781701e-03 0.017282072
## opendata                  5.443413e-02 0.076530312
## citation                  7.206905e-02 0.094131980
## vignette_r                2.615708e-02 0.051737694
## title_r:vignette_r       -2.691420e-02 0.018189004
## institution_r:vignette_r  4.784840e-03 0.056741951
## confirmation:vignette_r  -4.331936e-05 0.044904861
## quality:vignette_r       -4.195257e-02 0.003099084
## prestige:vignette_r      -2.646292e-02 0.018508058
## opendata:vignette_r      -4.029999e-02 0.004735656
## citation:vignette_r      -2.492316e-02 0.020093844
```

Bootstrapped confidence intervals for random effects

```
set.seed(42)
CIboot <- confint(mod_inter, method = 'boot', nsim = 2000, oldNames=FALSE)
```

```
## Computing bootstrap confidence intervals ...
```

```
##
## 3 message(s): boundary (singular) fit: see help('isSingular')
## 9 warning(s): Model failed to converge with max|grad| = 0.00200658 (tol = 0.002, component 1) (and o
```

```
CIboot
```

```
##                                   2.5 %      97.5 %
## sd_(Intercept)|CASE       0.1123940140 0.139180601
## sd_(Intercept)|item       0.0089330547 0.026390274
## sigma                     0.1949707494 0.202995754
## (Intercept)               0.5488690549 0.588383702
## title_r                  -0.0139889404 0.007795810
## institution_r            -0.0069018879 0.018473829
## confirmation              0.0462183963 0.068027346
## quality                   0.1993913193 0.221579503
## prestige                 -0.0050088385 0.017848765
## opendata                  0.0545097402 0.076256267
## citation                  0.0723406602 0.094088768
## vignette_r                0.0261577049 0.052035306
## title_r:vignette_r       -0.0257924711 0.019791348
```

```
## institution_r:vignette_r  0.0046742282 0.056607644
## confirmation:vignette_r   0.0006539491 0.044992895
## quality:vignette_r       -0.0420559106 0.002113575
## prestige:vignette_r      -0.0258712795 0.018632604
## opendata:vignette_r      -0.0396600660 0.004903286
## citation:vignette_r      -0.0249436696 0.020542611
```

Conditional and marginal R^2

```
r2(mod_inter)
```

```
## # R2 for Mixed Models
##
##   Conditional R2: 0.443
##      Marginal R2: 0.215
```

ICC

```
icc(mod_inter)
```

```
## # Intraclass Correlation Coefficient
##
##     Adjusted ICC: 0.290
##   Unadjusted ICC: 0.228
```

With random slopes Removed random slopes for vignette_r, confirmation, title_r, institution_r, and prestige to achieve convergence and prevent singular fit

```
mod_rs <- lmer(quality.rating ~ title_r + institution_r + confirmation + quality + prestige + opendata +
summary(mod_rs)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: quality.rating ~ title_r + institution_r + confirmation + quality +
##     prestige + opendata + citation + vignette_r + title_r:vignette_r +
##     institution_r:vignette_r + confirmation:vignette_r + quality:vignette_r +
##     prestige:vignette_r + opendata:vignette_r + citation:vignette_r +
##     (1 + quality + opendata + citation | CASE) + (1 | item)
##    Data: data_long
##
## REML criterion at convergence: -2009.8
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -4.1231 -0.5183  0.0451  0.5837  5.2897
##
## Random effects:
##  Groups   Name        Variance  Std.Dev. Corr
##  CASE     (Intercept) 0.0162231 0.12737
##           quality     0.0227622 0.15087  -0.08
##           opendata    0.0073750 0.08588  -0.14  0.24
```

```
##           citation    0.0072468 0.08513  -0.13  0.08  0.44
##  item     (Intercept) 0.0004498 0.02121
##  Residual             0.0297422 0.17246
## Number of obs: 5035, groups:  CASE, 210; item, 24
##
## Fixed effects:
##                             Estimate Std. Error         df t value Pr(>|t|)
## (Intercept)                5.678e-01  1.020e-02  1.785e+02  55.648  < 2e-16 ***
## title_r                   -3.550e-03  5.072e-03  4.501e+03  -0.700  0.48399
## institution_r              2.121e-03  5.882e-03  4.536e+03   0.361  0.71840
## confirmation               5.588e-02  5.117e-03  4.560e+03  10.920  < 2e-16 ***
## quality                    2.074e-01  1.156e-02  2.078e+02  17.950  < 2e-16 ***
## prestige                   7.792e-03  5.069e-03  4.490e+03   1.537  0.12432
## opendata                   5.910e-02  7.789e-03  2.080e+02   7.588 1.07e-12 ***
## citation                   8.035e-02  7.744e-03  2.069e+02  10.376  < 2e-16 ***
## vignette_r                 3.904e-02  5.923e-03  4.544e+03   6.591 4.88e-11 ***
## title_r:vignette_r        -5.830e-04  1.043e-02  4.582e+03  -0.056  0.95541
## institution_r:vignette_r   3.361e-02  1.201e-02  4.576e+03   2.797  0.00517 **
## confirmation:vignette_r    1.918e-02  1.042e-02  4.583e+03   1.840  0.06584 .
## quality:vignette_r        -1.305e-02  1.023e-02  4.493e+03  -1.276  0.20210
## prestige:vignette_r        5.206e-03  1.038e-02  4.555e+03   0.502  0.61604
## opendata:vignette_r       -1.655e-02  1.028e-02  4.418e+03  -1.609  0.10762
## citation:vignette_r       -1.068e-02  1.027e-02  4.418e+03  -1.040  0.29854
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

##
## Correlation matrix not shown by default, as p = 16 > 12.
## Use print(x, correlation=TRUE)  or
##     vcov(x)        if you need it
```

Direct Rating of Quality Indicators

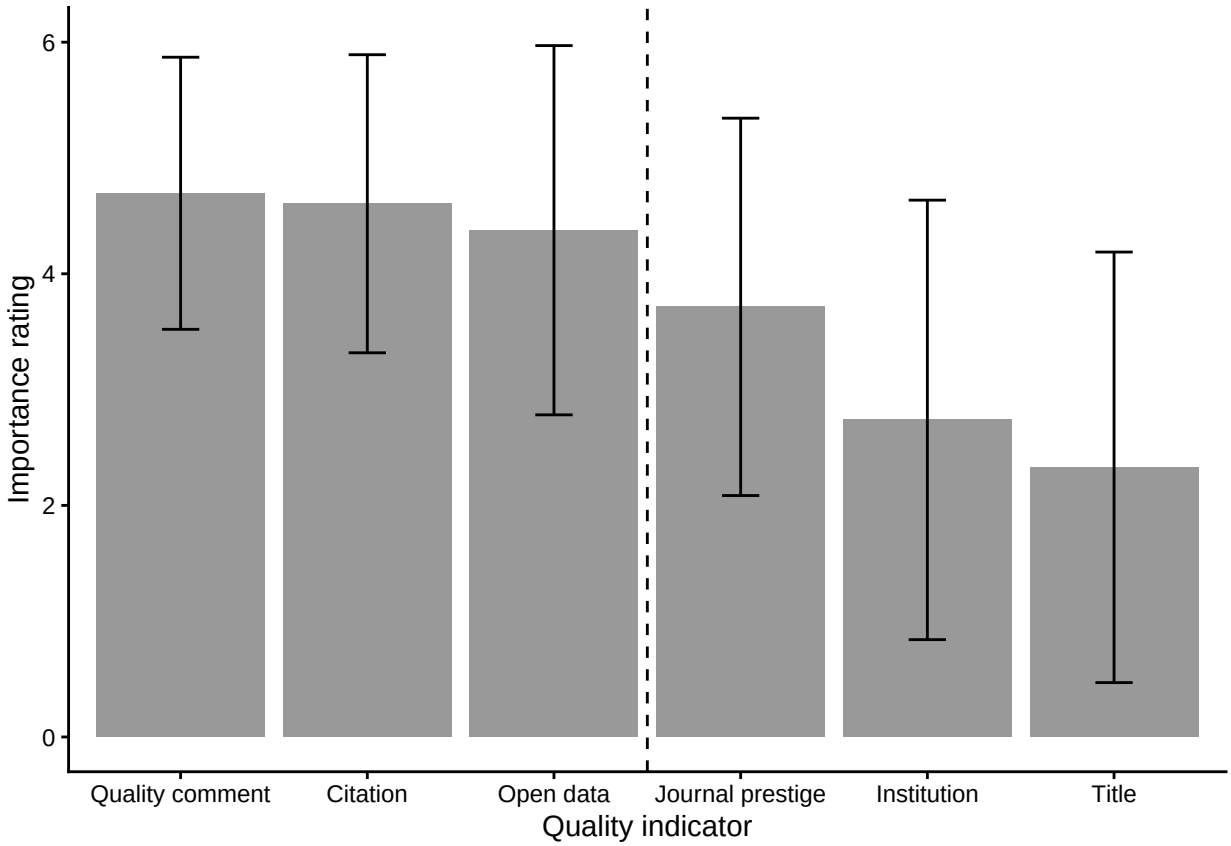
```
qi <- raw_data %>%
  select(CASE, quality.indicator_1:quality.indicator_6) %>%
  pivot_longer(cols = quality.indicator_1:quality.indicator_6,
               names_to = c("name", "indicator"),
               names_sep = "_") %>%
  select(-name) %>%
  mutate(indicator_txt = case_when(
    indicator == 1 ~ "title",
    indicator == 2 ~ "institution",
    indicator == 3 ~ "quality",
    indicator == 4 ~ "prestige",
    indicator == 5 ~ "opendata",
    indicator == 6 ~ "citation")) %>%
  mutate(value = value-1) %>%
  mutate(indicator_txt = factor(indicator_txt, levels = c("quality", "citation", "opendata", "prestige"

qi_avg <- qi %>%
  group_by(CASE, indicator_txt) %>%
  summarise(mean_ind = mean(value)) %>%
```

```
  group_by(indicator_txt) %>%
  summarise(mean = mean(mean_ind), se = sd(mean_ind)) %>%
  arrange(desc(mean))
```

```
## `summarise()` has grouped output by 'CASE'. You can override using the
## `.groups` argument.
```

```
p_qi <- ggplot(data = qi_avg, mapping = aes(x = indicator_txt, y = mean)) +
  geom_col(position = position_dodge(.9), fill = "grey60") +
  geom_errorbar(mapping = aes(ymin = mean-se, ymax = mean+se), width = .2, position = position_dodge(.9
  theme_classic() +
  ylim(c(0,6)) +
  labs(x = "Quality indicator",
       y = "Importance rating") +
  geom_vline(xintercept = 3.5, linetype = "dashed")
p_qi
```




```
# ggsave(p_qi, file = here("pilot2/plot_qi.svg"), height = 10, width = 15, limitsize = FALSE)
```

Analyze whether importance ratings are higher for selected quality indicators than for non-selected quality indicators

```
# contrast coding
qi <- qi %>%
```

```
  mutate(importance_contr = if_else(indicator_txt %in% c("Quality comment", "Open data", "citation"), 0

# mixed linear model
mod_importance <- lmer(value ~ importance_contr + (1|CASE), data = qi)
summary(mod_importance)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: value ~ importance_contr + (1 | CASE)
##    Data: qi
##
## REML criterion at convergence: 4931.1
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -2.6937 -0.6434  0.1462  0.6628  2.2484
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  CASE     (Intercept) 0.5081   0.7128
##  Residual             2.5640   1.6012
## Number of obs: 1260, groups:  CASE, 210
##
## Fixed effects:
##                    Estimate Std. Error        df t value Pr(>|t|)
## (Intercept)      3.941e+00  6.862e-02 2.334e+02   57.43   <2e-16 ***
## importance_contr 1.189e+00  9.569e-02 1.049e+03   12.43   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr)
## imprtnc_cnt 0.232
```

```
confint(mod_importance, method = "Wald", oldNames = FALSE)
```

```
##                         2.5 %   97.5 %
## sd_(Intercept)|CASE        NA       NA
## sigma                      NA       NA
## (Intercept)          3.806578 4.075565
## importance_contr     1.001731 1.376840
```

Bootstrapped confidence intervals for random effects

```
set.seed(42)
CIboot <- confint(mod_importance, method = 'boot', nsim = 2000, oldNames=FALSE)
```

```
## Computing bootstrap confidence intervals ...
```

```
CIboot
```

```
##                           2.5 %   97.5 %
## sd_(Intercept)|CASE 0.5729278 0.839430
## sigma               1.5345885 1.671156
## (Intercept)         3.8031224 4.069846
## importance_contr    1.0073199 1.381271
```

Conditional and marginal R^2

```
r2(mod_importance)
```

```
## # R2 for Mixed Models
##
##   Conditional R2: 0.243
##      Marginal R2: 0.093
```

ICC

```
icc(mod_importance)
```

```
## # Intraclass Correlation Coefficient
##
##     Adjusted ICC: 0.165
##   Unadjusted ICC: 0.150
```